

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are entered in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers)

2 8 3 9 7 6 8
N N N N N N N

Date of Birth (DD/MM/YYYY)

0 9 / 0 3 / 2 0 0 4
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Mark with an [X] which question you have completed on this sheet.
Cross one box only.

☐ Q1 ☐ Q2 ☐ Q3 ☒ Q4 ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE ☐ DO NOT USE

Q4.

You may use both sides but only answer **ONE** question per sheet

(a).

i. The bilinear substitution for s :

$$s = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}$$

From $H(s) = \frac{1}{s+1}$, and $T=1s$,

$$H(z) = \frac{1}{\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} + 1} = \frac{z^{-1}+1}{-z^{-1}+3} = -\frac{z^{-1}+1}{z^{-1}-3}$$

∴ For bilinear transform, it can map all the left plane

of s domain into the unit circle in the z -plane,

which can ensure the stability.

As for the matched z -transform, it can not ensure that.

ii.

$$H(z) = -\frac{z^{-1}+1}{z^{-1}-3} \Rightarrow Y(z) = \frac{1}{3} Y(z) + \frac{1}{3} X(z) \Rightarrow Y(z) = \frac{1}{2} X(z)$$

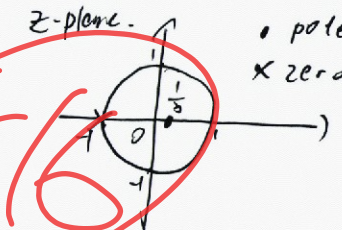
therefore, $a_1 = \frac{1}{3}$

$b_0 = \frac{1}{3}, b_1 = \frac{1}{3}$

(b) There is 1 zero: $z_1 = -1$.

There is 1 pole: $p_1 = \frac{1}{3}$

It is stable because every pole is located in the unit circle ($\frac{1}{3} < 1$).



(c)

(i) The pole near the unit circle can enhance the response at correspondant frequency while the zero near the unit circle can attenuate the response at the correspondant frequency.

(ii) Because the pole is nearer to the unit circle, the response will be more sharp.

$$(iii) H(z) = \frac{1-2\cos(\omega_0)z^{-1}+z^{-2}}{1-1.4\cos(\omega_0)z^{-1}+(0.7)^2z^{-2}}$$

TOTAL 23

Session

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Exam

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Q Continued

Its frequency response is

$$\begin{aligned}
 H(e^{j\omega}) &= \frac{1 - 2\cos\omega_0 \cdot e^{-j\omega} + e^{-2j\omega}}{1 - 1.4\cos\omega_0 \cdot e^{-j\omega} + 0.7^2 \cdot e^{-2j\omega}} \\
 &= \frac{e^{j\omega}(e^{j\omega} + e^{-j\omega}) - 2\cos\omega_0 \cdot e^{-j\omega}}{e^{j\omega}(e^{j\omega} + e^{-j\omega}) \cdot (0.7)^2 + 0.51 - 1.4\cos\omega_0 \cdot e^{-j\omega}} \\
 &= \frac{2\cos\omega - 2\cos\omega_0}{2\cos\omega \cdot (0.7)^2 - 1.4\cos\omega_0 + 0.51} e^{j\omega}
 \end{aligned}$$

$$|H(e^{j\omega})| = \frac{|\cos\omega - \cos\omega_0|}{\left|\cos\omega - \frac{\cos\omega_0}{0.7}\right|} \cdot \frac{1}{(0.7)^2}$$

It has one zero at $\omega = \omega_0 = 0.3\pi$.

one pole at $\omega = \arccos\left(\frac{\cos\omega_0}{0.7}\right) = 0.574$.

So it is a bandstop filter (notch filter) that can filter out frequency around $\omega = 0.3\pi$.

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